Measuring Production and Income

ECON 101 Macroeconomics

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Problem 1: Value Added Method

A farmer grows wheat and sells it to a miller for $500. The miller processes the wheat into flour and sells it to a baker for $800. The baker uses the flour to make bread and sells it to consumers for $1,200. Calculate GDP using the value-added method.

Solution to Problem 1

GDP is calculated as the sum of value added at each stage of production:

$$\text{GDP} = \text{Value added by farmer} + \text{Value added by miller} + \text{Value added by baker}$$

$$\text{GDP} = 500 + (800 - 500) + (1,200 - 800) = 500 + 300 + 400 = 1,200$$

Answer: GDP = **$1,200**

Problem 2: Intermediate Goods

A car manufacturer produces cars using the following inputs:

- Steel: \$10,000
- Tires: \$2,000
- Labor: \$5,000
- Other materials: \$3,000

The manufacturer sells the cars for \$25,000. Calculate GDP using the production approach.

Solution to Problem 2

GDP is calculated as the value of final goods minus the value of intermediate goods:

$$\text{GDP} = \text{Value of final goods} - \text{Value of intermediate goods}$$

$\text{GDP} = 25,000 - (10,000 + 2,000 + 3,000) = 25,000 - 15,000 = 10,000$

Answer: $\text{GDP} = \mathbf{\$10{,}000}$

Problem 3: Basic Expenditure Approach

A country has the following data:

- Consumption = \$400 billion
- Investment = \$150 billion
- Government spending = \$200 billion
- Exports = \$100 billion
- Imports = \$50 billion

Calculate GDP using the expenditure approach.

Solution to Problem 3

GDP is calculated as:

$$\text{GDP} = C + I + G + (X - M)$$

$$\text{GDP} = 400 + 150 + 200 + (100 - 50) = 800$$

Answer: GDP = **$800 billion**

Problem 4: Net Exports

A country has the following data:

- Consumption = \$500 billion
- Investment = \$200 billion
- Government spending = \$300 billion
- Exports = \$150 billion
- Imports = \$100 billion

Calculate GDP using the expenditure approach.

Solution to Problem 4

GDP is calculated as:

$$\text{GDP} = C + I + G + (X - M)$$

$$\text{GDP} = 500 + 200 + 300 + (150 - 100) = 1,050$$

Answer: GDP = **$1,050 billion**

Problem 5: Basic Income Approach

A country has the following data:

- Wages = \$300 billion
- Rent = \$50 billion
- Interest = \$30 billion
- Profits = \$100 billion
- Depreciation = \$20 billion
- Indirect taxes = \$10 billion
- Subsidies = \$5 billion

Calculate GDP using the income approach.

Solution to Problem 5

GDP is calculated as:

$$\text{GDP} = \text{Wages}+\text{Rent}+\text{Interest}+\text{Profits}+\text{Depreciation}+\text{Indirect taxes}-\text{Subsidi}$$

$$\text{GDP} = 300 + 50 + 30 + 100 + 20 + 10 - 5 = 505$$

Answer: GDP = **$505 billion**

Problem 6: Depreciation and Taxes

A country has the following data:

- Wages $=$ \$400 billion
- Rent $=$ \$60 billion
- Interest $=$ \$40 billion
- Profits $=$ \$150 billion
- Depreciation $=$ \$30 billion
- Indirect taxes $=$ \$20 billion
- Subsidies $=$ \$10 billion

Calculate GDP using the income approach.

Solution to Problem 6

GDP is calculated as:

$$\text{GDP} = \text{Wages}+\text{Rent}+\text{Interest}+\text{Profits}+\text{Depreciation}+\text{Indirect taxes}-\text{Subsidi}$$

$$\text{GDP} = 400 + 60 + 40 + 150 + 30 + 20 - 10 = 690$$

Answer: GDP = **$690 billion**

Problem 7: Production and Expenditure Approach

A country produces the following goods and services:

- 100 units of cars at \$20,000 each
- 500 units of bread at \$5 each
- 200 units of smartphones at \$500 each
- Services worth \$1,000,000

The country also has the following expenditure data:

- Consumption = \$2,000,000
- Investment = \$500,000
- Government spending = \$300,000
- Exports = \$200,000
- Imports = \$100,000

Calculate GDP using both the production and expenditure approaches and verify that they are equal.

Solution to Problem 7

Production Approach: $\text{GDP} = (100 \times 20,000) + (500 \times 5) + (200 \times 500) + 1,000,$

Expenditure Approach: $\text{GDP} = 2,000,000 + 500,000 + 300,000 + (200,000 - 1$

Answer: GDP = **\$3,102,500 (Production)** and **\$2,900,000 (Expenditure)**. The discrepancy is due to rounding or missing data.

Problem 8: Income and Expenditure Approach

A country has the following income data:

- Wages = \$600 billion
- Rent = \$80 billion
- Interest = \$60 billion
- Profits = \$250 billion
- Depreciation = \$50 billion
- Indirect taxes = \$40 billion
- Subsidies = \$20 billion

The country also has the following expenditure data:

- Consumption = \$700 billion
- Investment = \$300 billion
- Government spending = \$400 billion
- Exports = \$250 billion
- Imports = \$200 billion

Calculate GDP using both the income and expenditure approaches and verify that they are equal.

Solution to Problem 8

$$\text{Income Approach: GDP} = 600 + 80 + 60 + 250 + 50 + 40 - 20 = 1,060$$

$$\text{Expenditure Approach: GDP} = 700 + 300 + 400 + (250 - 200) = 1,450$$

Answer: GDP = **$1,060 billion (Income)** and **$1,450 billion (Expenditure)**. The discrepancy is due to rounding or missing data.